



Course Title: Investment Analysis and Portfolio Management

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Masters of Business Administration
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Company Overview and Financial Data

JPMorgan Chase & Co. (JPM)

Company Overview

- **Industry:** Financial Services
- **Headquarters:** New York City, New York, USA
- **Founded:** 2000 (via merger; historical roots date back to 1799)
- **CEO:** Jamie Dimon (since 2006)
- **Employees:** Over 317,000 (as of 2024)
- **Total Assets:** Approximately \$4.4 trillion (as of March 31, 2025)
- **Market Capitalization:** Approximately \$707 billion (early 2025)
- **Stock Price:** \$253.47 (as of May 9, 2025)

JPMorgan Chase & Co. is the largest bank in the United States by total assets and one of the most influential financial institutions globally. The firm offers a comprehensive range of services, including investment banking, commercial banking, asset management, and consumer finance. It operates through several key business segments:

Consumer & Community Banking: Provides services such as credit cards, mortgages, and auto loans under the Chase brand.

Corporate & Investment Bank: Offers investment banking, market-making, prime brokerage, and treasury and securities services.

Commercial Banking: Delivers financial solutions, including lending and treasury services, to small and mid-sized businesses, municipalities, and nonprofits.

Asset & Wealth Management: Provides investment management across all major asset classes, including equities, fixed income, alternatives, and money market funds.

The bank serves millions of customers and many of the world's most prominent corporate, institutional, and government clients. Its global operations span over 100 markets.

Recent Financial Performance

JPMorgan Chase reported record-breaking financial results for the year 2024:

Revenue: Over \$177 billion, marking a significant increase from the previous year.

Net Income: \$58.5 billion, the highest annual profit in the bank's history.

Earnings Per Share (EPS): \$19.75.

In the first quarter of 2025, the bank continued its strong performance:

Earnings Per Share (Q1 2025): \$5.07.

Revenue (Q1 2025): \$45.31 billion.

Strategic Developments

Artificial Intelligence Integration: JPMorgan has significantly invested in artificial intelligence (AI) to enhance its services and operations. The bank's AI tools, such as the "Coach AI," have improved client interactions by providing personalized investment insights. These tools contributed to a 20% increase in asset and wealth management sales between 2023 and 2024. Additionally, AI-driven efficiencies have led to cost savings of nearly \$1.5 billion through improvements in fraud prevention.

Expansion into Private Credit: In a strategic move to diversify its revenue streams, JPMorgan has allocated \$50 billion to direct lending in the private credit market, targeting private equity-backed companies. This initiative positions the bank as a significant player in the growing \$2 trillion private credit asset class.

Leadership and Compensation: CEO Jamie Dimon received a total compensation of \$39 million for 2024, reflecting the bank's strong performance. Under his leadership, JPMorgan has navigated complex economic landscapes and maintained its position as a global financial leader.

Outlook

Looking ahead, JPMorgan Chase aims to continue leveraging technology and strategic investments to drive growth. The bank's focus on AI integration, expansion into new markets, and commitment to operational excellence positions it well to adapt to evolving financial landscapes and maintain its leadership in the global banking sector.

NVIDIA Corporation (NVDA)

Company Overview

- **Industry:** Semiconductors / Technology
- **Headquarters:** Santa Clara, California, USA
- **Founded:** 1993
- **CEO:** Jensen Huang
- **Employees:** Approximately 36,000 (as of 2024)
- **Market Capitalization:** Approximately \$2.7 trillion (as of May 2025)
- **Stock Price:** \$117.37 (as of May 9, 2025)

NVIDIA Corporation is a global leader in visual computing, high-performance computing, and artificial intelligence technologies. Originally established as a graphics hardware company focused on gaming, NVIDIA has expanded far beyond its original scope and is now at the forefront of AI, data centers, and autonomous systems.

The company operates through several key business segments and product platforms:

Gaming: Home to the GeForce GPU product line, which powers PC gaming, esports, and immersive virtual reality experiences. This remains a core revenue driver.

Data Center: Powered by NVIDIA A100, H100, and the new Blackwell B100/B200 chips. These GPUs are designed for AI model training, inference, and cloud computing workloads.

Professional Visualization: Includes NVIDIA RTX graphics cards for designers, engineers, and content creators across industries such as architecture, media, and manufacturing. **Automotive:** Through its NVIDIA DRIVE platform, the company offers solutions for autonomous vehicles, including hardware, AI software, and simulation technology.

OEM & IP: Covers licensing and chip production for third-party applications, including embedded devices and IoT platforms.

Networking (via Mellanox acquisition): Provides high-performance interconnect solutions for data centers, AI workloads, and high-speed cloud computing.

Recent Financial Performance

NVIDIA has demonstrated remarkable financial growth in recent years:

Fiscal Year 2024 Revenue: \$60.9 billion, up 126% year-over-year.

Fiscal Year 2025 Revenue: \$130.5 billion, up 114% year-over-year.

Q4 Fiscal 2025 Revenue: \$39.3 billion, up 78% year-over-year.

This explosive growth is attributed to its dominant position in the AI hardware market and surging global demand for generative AI technologies.

Strategic Developments

AI and Data Center Expansion: NVIDIA's rapid growth has been fueled by the surge in demand for AI and data center solutions. The company's GPUs are widely used in training deep learning models, driving the AI boom.

Blackwell Architecture: Introduced in 2024, this next-generation GPU line offers transformative computational performance and energy efficiency for AI and large-scale cloud deployments.

Global Expansion: NVIDIA announced its first R&D center in Vietnam in 2024, expanding its global innovation footprint.

Leadership Compensation: CEO Jensen Huang received \$49.87 million in total compensation for 2025, reflecting his key role in NVIDIA's exponential growth and leadership in AI.

Outlook

Looking ahead, NVIDIA aims to continue leveraging its technological advancements and strategic investments to drive growth. The company's focus on AI integration, expansion into new markets, and commitment to innovation positions it well to adapt to evolving technological landscapes and maintain its leadership in the AI-driven semiconductor industry.

The Coca-Cola Company (KO)

Company Overview

- Industry: Non-Alcoholic Beverages
- Headquarters: Atlanta, Georgia, USA
- Founded: 1886

- CEO: James Quincey
- Employees: Approximately 86,200 (as of 2024)
- Market Capitalization: Approximately \$311.6 billion (as of May 2025)
- Stock Price: \$71.17 (as of May 9, 2025)

Company Overview

The Coca-Cola Company is a global leader in the non-alcoholic beverage industry, renowned for its flagship product, Coca-Cola. Founded in 1886, the company has expanded its portfolio to include over 200 brands, encompassing sparkling soft drinks, water, juices, teas, coffees, and energy drinks. Its products are available in more than 200 countries worldwide, serving millions of consumers daily.

The company operates through a franchise model, with bottling partners handling the manufacturing, packaging, and distribution of its products. This model allows Coca-Cola to maintain a broad global presence while focusing on brand development, marketing, and innovation.

Recent Financial Performance

In 2024, Coca-Cola reported:

- **Revenue:** \$47.1 billion, up 2.9% from FY 2023.
- **Net Income:** \$10.6 billion, flat compared to FY 2023.
- **Earnings Per Share (EPS):** \$2.47, slightly down from \$2.48 in FY 2023.
- **Operating Income:** \$9.99 billion, a decline from \$11.31 billion in 2023, primarily due to higher operating charges.

The North America segment contributed significantly to revenue, accounting for 40% of total revenue in 2024.

Strategic Developments

Health-Conscious Product Innovation: Coca-Cola has been expanding its portfolio to include healthier beverage options, such as low-sugar and functional drinks, in response to changing consumer preferences.

Sustainability Initiatives: The company is committed to sustainability, aiming to reduce its carbon footprint and increase the use of recyclable materials in its packaging.

Digital Transformation: Coca-Cola is investing in digital technologies to enhance consumer engagement and streamline its operations.

Outlook

Looking ahead, Coca-Cola aims to continue its growth by focusing on product innovation, sustainability, and digital transformation. The company's strong brand portfolio and global presence position it well to adapt to evolving consumer preferences and market dynamics.

2-Year Market Price Analysis (April 30, 2023 – April 30, 2025)

1. JPMorgan Chase & Co. (JPM)

April 30, 2023 Closing Price: \$136.94

April 30, 2025 Closing Price: \$253.47

2-Year Price Change: +\$116.53 (+85.1%)

JPMorgan Chase's stock has demonstrated robust growth over the past two years. In April 2023, the stock was priced at \$136.94, and by April 2025, it had risen to \$253.47—marking an 85.1% increase. This performance highlights the bank's resilience, strong earnings reports, and successful navigation of macroeconomic conditions. The upward trajectory also reflects investor confidence in its diverse business segments and global leadership in financial services.

2. NVIDIA Corporation (NVDA)

April 30, 2023 Closing Price: \$27.74

April 30, 2025 Closing Price: \$117.37

2-Year Price Change: +\$89.63 (+323.2%)

NVIDIA's stock has seen extraordinary growth over the last two years. Starting at \$27.74 in April 2023, it surged to \$117.37 by April 2025—representing an impressive 323.2% increase. This explosive growth is fueled by rising demand for NVIDIA's GPUs in AI, cloud computing, and

autonomous systems. Investor sentiment has been significantly boosted by NVIDIA’s leading position in the generative AI space and exponential data center revenues.

3. The Coca-Cola Company (KO)

April 30, 2023 Closing Price: \$63.99

April 30, 2025 Closing Price: \$71.17

2-Year Price Change: +\$7.18 (+11.2%)

The Coca-Cola Company has shown steady but moderate stock price growth. From \$63.99 in April 2023, the stock increased to \$71.17 by April 2025—an 11.2% gain. The company’s consistent dividends, global brand recognition, and strong consumer loyalty have helped maintain its value. While not as explosive as tech or banking stocks, Coca-Cola remains a stable investment in the consumer staples sector.

Summary Table

Company	April 30, 2023 Price	April 30, 2025 Price	2-Year Change
JPMorgan Chase (JPM)	\$136.94	\$253.47	+85.1%
NVIDIA (NVDA)	\$27.74	\$117.37	+323.2%
Coca-Cola (KO)	\$63.99	\$71.17	+11.2%

1. JPMorgan Chase (JPM)

i. Returns

- **Average Arithmetic Return:** 0.0758% per day
- **Geometric Mean Return:** 0% (may be a data issue—likely very low)
- JPM provides a moderate return. The arithmetic average suggests modest gains on most trading days.

ii. Risk (Standard Deviation)

- Daily Std. Deviation: 1.5743

- Variance: 0.0248%

Moderate volatility. Less risky than NVIDIA, more than KO. Suitable for balanced portfolios.

iii. Correlation & Covariance

Positive with KO and market due to being in financial sector. As a financial stock, JPM typically moves with macroeconomic factors and the market index.

iv. Market Risk Premium & Factor Analysis

From **Fama-French** **model:**

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.168769
R Square	0.028483
Adjusted R Square	0.023274
Standard Error	0.015671
Observations	751

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	4	0.005371	0.001343	5.467823	0.000242
Residual	746	0.183211	0.000246		
Total	750	0.188583			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	-0.00224	0.002435	-0.92045	0.357635	-0.00702	0.002539	-0.00702	0.002539
X Variable 1	-9.6E-05	0.000891	-0.10774	0.914229	-0.00184	0.001652	-0.00184	0.001652
X Variable 2	-0.00453	0.001558	-2.90511	0.00378	-0.00759	-0.00147	-0.00759	-0.00147
X Variable 3	0.00174	0.001381	1.259679	0.208179	-0.00097	0.004452	-0.00097	0.004452
X Variable 4	-0.72772	0.214914	-3.38612	0.000746	-1.14963	-0.30582	-1.14963	-0.30582

- $R^2 = 2.85\%$

- Adjusted $R^2 = 2.33\%$

Market factors explain very little of JPM's return. Most movement is stock-specific (idiosyncratic risk), possibly due to banking sector events.

v. **Jensen's** **Alpha** **(Portfolio-level)**

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.614295
R Square	0.377358
Adjusted R Square	0.376527
Standard Error	0.012521
Observations	751

<i>ANOVA</i>					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	0.071163	0.071163	453.9388	4.14E-79
Residual	749	0.117419	0.000157		
Total	750	0.188583			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	-0.00102	0.00063	-1.62322	0.104963	-0.00226	0.000214	-0.00226	0.000214
X Variable 1	0.864999	0.040599	21.30584	4.14E-79	0.785297	0.9447	0.785297	0.9447

This affects JPM if included in the portfolio.

Low R^2 suggests alpha plays a bigger role than beta; active management may yield better results.

2. Coca-Cola (KO)

i. Returns

- **Average Arithmetic Return:** 0.0229% per day
- Geometric Return: 1.6735% annually (likely misaligned—needs clarification)
- Interpretation: Low return, but very stable. Reflects defensive nature of consumer staples.

ii. Risk (Standard Deviation)

- Std. Deviation: 0.984%
- Variance: 0.0097%

Lowest risk among the three. Very stable stock, excellent for preserving capital or conservative investors.

iii. Correlation & Covariance

Likely positive with market but weaker than JPM.

KO is less sensitive to economic cycles and market swings due to consistent consumer demand.

iv. Market Risk Premium & Factor Analysis

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.298503
R Square	0.089104
Adjusted R Square	0.08422
Standard Error	0.009849
Observations	751

<i>ANOVA</i>					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	4	0.007079	0.00177	18.24345	2.61E-14
Residual	746	0.072363	9.7E-05		
Total	750	0.079441			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	0.001665	0.00153	1.08808	0.276911	-0.00134	0.004669	-0.00134	0.004669
X Variable 1	0.000178	0.00056	0.318758	0.75	-0.00092	0.001277	-0.00092	0.001277
X Variable 2	-0.00148	0.000979	-1.50795	0.131992	-0.0034	0.000446	-0.0034	0.000446
X Variable 3	0.000385	0.000868	0.443734	0.657363	-0.00132	0.002089	-0.00132	0.002089
X Variable 4	-1.13198	0.135066	-8.38099	2.6E-16	-1.39714	-0.86683	-1.39714	-0.86683

From Fama-French KO sheet:

- $R^2 = 8.91\%$
- Adjusted $R^2 = 8.42\%$

A bit more aligned with market factors than JPM, but still largely independent. Stability reduces exposure to risk factors.

3. NVIDIA (NVDA)

i. Returns

- Average Arithmetic Return: 0.2623% per day — highest among the three
- Geometric Return: -5.7456% (likely an error due to volatility or log-scale misalignment)

Offers very high return potential, but questionable sustainability or consistent profitability.

ii. Risk (Standard Deviation)

- Daily Std. Deviation: 3.4810%
- Variance: 0.1212%

Interpretation: Highly volatile and speculative. Excellent for aggressive investors but not suitable for risk-averse profiles.

iii. Correlation & Covariance

Likely low or negatively correlated with KO.

Offers potential diversification if mixed with stable stocks like KO.

iv. **Market** **Risk** **Premium** **&** **Factor** **Analysis**

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.094291
R Square	0.008891
Adjusted R Square	0.003577
Standard Error	0.034758
Observations	751

<i>ANOVA</i>					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	4	0.008085	0.002021	1.673019	0.154321
Residual	746	0.901245	0.001208		
Total	750	0.90933			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	-0.00347	0.005401	-0.64176	0.521226	-0.01407	0.007136	-0.01407	0.007136
X Variable 1	0.004103	0.001975	2.077085	0.038135	0.000225	0.007981	0.000225	0.007981
X Variable 2	0.002482	0.003456	0.718187	0.472867	-0.0043	0.009268	-0.0043	0.009268
X Variable 3	0.002317	0.003064	0.756258	0.449734	-0.0037	0.008331	-0.0037	0.008331
X Variable 4	-0.47685	0.476661	-1.00041	0.317438	-1.41261	0.458902	-1.41261	0.458902

From Fama-French NVIDIA sheet:

- $R^2 = 0.89\%$
- Adjusted $R^2 = 0.35\%$

Almost completely uncorrelated with market factors. Returns driven by firm-specific innovation and tech sentiment, not macro indicators.

4. Market (S&P 500)

Average Daily Return: 0.0338%

- Std. Deviation: 1.1022%
- Annualized Market Return: 13.13%

Jensen's Alpha (Portfolio-Level, applicable if these stocks are part of the portfolio)

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.70901
R Square	0.502695
Adjusted R Square	0.502031
Standard Error	0.024571
Observations	751

<i>ANOVA</i>					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	0.457115	0.457115	757.1166	1E-115
Residual	749	0.452215	0.000604		
Total	750	0.90933			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	0.015019	0.001236	12.14989	3.94E-31	0.012592	0.017446	0.012592	0.017446
X Variable 1	2.192303	0.079674	27.51575	1E-115	2.035891	2.348715	2.035891	2.348715

From regression sheet:

$R^2 = 37.7\%$

Suggests market returns explain a reasonable part of the portfolio's variance. Positive alpha implies active management added value beyond market expectations.

Summary Table

Metric	JPM	KO	NVIDIA
Avg Daily Return	0.0758%	0.0229%	0.2623%
Std. Deviation	1.5743%	0.984%	3.4810%
R ² (Fama-French Regression)	2.85%	8.91%	0.89%
Risk Level	Moderate	Low	Very High
Return Potential	Medium	Low	High
Best For	Balanced	Conservative.	Aggressive